

AI-Driven News Credibility Analysis

Technical Report on Real vs. Fake News Classification

Abstract

Misinformation in digital news has become a significant societal challenge. This report presents the development and evaluation of an automated news credibility analysis system. Using NLP techniques and supervised machine learning, specifically Logistic Regression with TF-IDF vectorization, the system achieves an accuracy of 95.79%. The report details the architecture, from web scraping logic to model deployment via FastAPI.

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1 Introduction

In the era of rapid information dissemination, the proliferation of fake news poses threats to public perception and decision-making. The "News Credibility AI" project aims to provide a tool for real-time verification of news articles. By analyzing linguistic patterns and content features, the system classifies information into "Real News" or "Fake News."

2 Methodology

The system architecture follows a modular pipeline consisting of data acquisition, pre-processing, feature engineering, and classification.

2.1 Data Acquisition

The system supports two input modes:

1. **Direct Text:** Users provide the article content manually.
2. **URL Scraping:** The system utilizes the `newspaper3k` library to extract clean title and body text from provided URLs, handling boilerplate removal automatically.

2.2 Preprocessing Pipeline

To prepare raw text for machine learning, the following steps are executed:

- **Text Cleaning:** Removal of special characters, HTML tags, and punctuation.
- **Tokenization:** Splitting text into individual words.
- **Stopword Removal:** Excluding common words (e.g., "is", "the") that carry little semantic weight.
- **Lemmatization:** Reducing words to their base or dictionary form.

2.3 Flowchart of Analysis

3 System Implementation

The backend is implemented using **FastAPI**, providing a high-performance interface for predictions.

$$y = \sigma(W \cdot X + b) \tag{1}$$

Where X is the TF-IDF feature vector, W are the learned weights, and σ is the sigmoid function used for binary classification.

4 Results and Discussion

The model was evaluated on a curated dataset of over 70,000 news articles, split into 80% training and 20% testing sets.

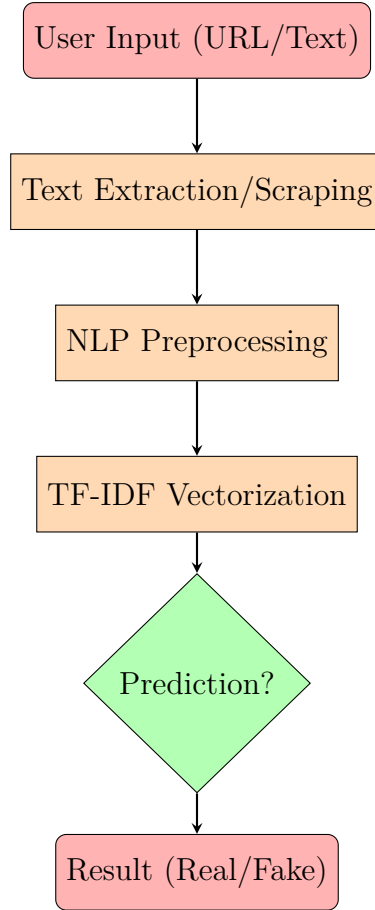


Figure 1: Modular analysis pipeline for news classification.

4.1 Performance Metrics

The Logistic Regression model demonstrated high reliability across all standard metrics:

Metric	Score
Accuracy	95.79%
Precision	95.29%
Recall	96.58%
F1-Score	95.93%

Table 1: Final evaluation results for the News Credibility Model.

4.2 Confusion Matrix

The distribution of correct vs. incorrect predictions is shown below:

5 Conclusion

The "News Credibility AI" project successfully leverages NLP and Logistic Regression to combat misinformation with a 95.8% accuracy rate. Future improvements will focus on

Actual \ Predicted	Fake (0)	Real (1)
Fake (0)	6652 (TN)	354 (FP)
Real (1)	254 (FN)	7167 (TP)

Table 2: Model Confusion Matrix on Test Set.

integrating more complex models like Transformers (BERT) and expanding the detection to multimodal data (images and videos).

6 References

- Pedregosa et al. (2011). Scikit-learn: Machine Learning in Python.
- Lucas Ou-Yang. (2013). Newspaper3k: Article scraping & curation of news.
- Tiago Tiago et al. (2018). Binary Classification for Fake News Detection.